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Biochemistry & Biophysics

Angiotensin II Enhances Activity and Flexibility in Murine CD8 T Cell Immunometabolism

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Purpose. Cytotoxic (CD8) T cells play a crucial role in adaptive immunity against intracellular pathogens and tumors by killing infected and mutated cells. Persistent antigen exposure, however, can induce an “exhausted” CD8 phenotype characterized by inhibitory receptors, impaired cytokine production, and poor proliferative potential and survival. As such, exhaustion hinders an effective immune response against cancerous cells, leading to worse disease prognoses and less effective immunotherapies. In this study, we examined CD8 exhaustion in response to elevated angiotensin II (Ang II) levels, a phenotype implicated in many malignancies. We expected the CD8s of mice with elevated levels of Ang II to exhibit increased exhaustion with a less active and robust metabolic profile, due to the bidirectional relationship between hypertension and chronic inflammation.

Methods. We implanted subcutaneous osmotic pumps containing Ang II or vehicle into 21-day-old male C57BL/6 wild-type mice, incubating for 28 days. We then ran SCENITH analysis of CD8s isolated from harvested mouse spleens. SCENITH (Single Cell ENergetic metabolism by profilIng Translation inHibition) incorporates flow cytometry to reveal various cell populations’ metabolic profiles.

Results. The mice of the Ang II treatment group (which were shown to be hypertensive from characteristic fibrosis evident in renal and aortic histology slides) had bulk CD8s that, despite elevated levels of exhaustion markers, exhibited a decreased mitochondrial dependence (MD), a increased glycolytic capacity (GC), and an increased fatty acid oxidation/amino acid oxidation capacity (FAO/AAO cap). When we compared just the exhausted CD8s of the two treatment groups, once again, the hypertensive CD8s exhibited a decreased glycolytic dependence (GD), a decreased MD, an increased GC, and an increased FAO/AAO cap. This was the trend for both examined exhausted CD8 populations.

Conclusion. Even though Ang II is associated with increased CD8 exhaustion, our results demonstrate its role in enhancing metabolic activity and flexibility in CD8 T cells, especially in the exhausted subsets. This reveals the need for further elucidation and unraveling of the various effects of Ang II, including its role exacerbating exhaustion, acting as a co-stimulatory molecule in CD8 activation, promoting tumor growth and metastasis, and as a mediator of hypertension and inflammation, etc.

Biochemical Characterization of Human LIG1 Syndrome Variants as Possible Players in Cancer Expression

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Purpose. Human DNA ligase 1 (LIG1) uses ATP and magnesium in the final steps of nucleotide excision repair, long-patch base excision repair, and normal DNA replication. Mutation of this enzyme can lead to human LIG1 syndrome, characterized by immunodeficiency and developmental delays. There is growing evidence for the involvement of R305Q, R771G, and R771W LIG1 syndrome mutations in genome instability and non-specific patient cancers. Determining whether rare LIG1 mutants increase cell mutagenesis by shifting enzyme fidelity would underscore LIG1's involvement in cancer formation and highlight a relatively new therapeutic target for cancer prevention.

Methods. Steady-state kinetics competition assays were run before using urea polyacrylamide gel electrophoresis (PAGE) for product separation and quantification. Assays were run at 37°C and quenched in FLS. An analysis of initial turnover rates allowed me to look at how well these different LIG1 mutants discriminated against sealing oxidized (damaged) DNA single-stranded breaks and preferentially closed the backbone of healthy canonical DNA.

Results. All mutants were able to better discriminate against G:A mismatches compared to WT LIG1. As cofactor increased, fidelity decreased for all LIG1 species except R771G.

Conclusion. The LIG1 variants tested all have higher fidelity than WT using a 3' 8oxoG:A pair. These mutants appear unlikely to increase cancer risk than WT LIG1. While R771W and R305Q are sensitive to increasing free [Mg²⁺], R771G appears resistant to increasing free cofactor. This outcome favors a LIG1 syndrome model in which longer-living single-stranded breaks are a source of mutations.

Impact of Carbohydrate Sources On Prophage Induction In *Phocaeicola vulgatus*

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Purpose. In a previous murine model of a stable synthetic gut community, a mobile genetic element (MGE) identified as a prophage transferred from the invading microbe *P. vulgatus* to a dominant community member. This transfer coincided with diet-dependent prophage induction, with increased copy number variation (CNV) observed in mice fed a standard diet but not a high-sugar diet. Here, we investigated how carbon source and concentration regulate prophage induction and lysis dynamics of *P. vulgatus* in vitro.

Methods. Follow-up in vitro experiments tested *P. vulgatus* growth in defined minimal media with various carbon sources such as D-glucose, L-arabinose, or rhamnogalacturonan-I (RG1) at concentrations ranging from 0.25–5 mg/mL.

Results. Growth curves showed that at 5 mg/mL, lysis did not occur despite entry into the stationary phase, suggesting alternative regulatory pathways limit activation under saturating carbon conditions. Lysis was most pronounced at 2.5 mg/mL, and induction occurred even at 0.25 mg/mL. Phase contrast and LIVE/DEAD microscopy confirmed reduced viability post-lysis. qPCR targeting phage excision (*attP*) revealed prophage induction in a small subset of cells, with CNV increases preceding lysis.

Conclusion. These findings support that prophage induction in *P. vulgatus* is carbon source and concentration-dependent, and regulated by mechanisms beyond simple growth phase entry. Future work will involve expanding qPCR to measure phage particles for better understanding of what goes on at each time point in the growth assay, curing *P. vulgatus* of the prophage to assess fitness effects, and performing colony-forming unit assays to complement OD measurements.

NCAM2 is required for gastrulation and body axis patterning during embryogenesis in *Nematostella*

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Purpose. Embryonic development is a fundamental biological process essential for life in all animals. Embryogenesis depends on the regulated expression of proteins that establish axial polarity and the adhesion of epithelial cells in the early embryo. The purpose of this study is to determine whether NCAM2 contributes to these processes, with the goal of uncovering a novel function for NCAM2 during embryonic development in the starlet sea anemone *Nematostella vectensis*.

Methods. Adult *N. vectensis* polyps were maintained at 17 °C in dark conditions, and gametes were obtained by induced spawning. Embryos were fertilized and cultured at 21 °C. NCAM2, -catenin, and GFP (control) were knocked down using shRNA electroporation. Embryos were fixed at late blastula (18 hpf) and gastrula (26 hpf) stages for phenotypic and molecular analyses. Protein localization was assessed by immunostaining for Cadherin-3 and phosphorylated myosin light chain, and staining with phalloidin and DAPI for F-actin and DNA. Gene expression of the transcription factors *snailA* and *brachyury* was assessed using HCR in situ hybridization. All samples were imaged using confocal microscopy (Zeiss LSM980). Structural predictions of NCAM2 were generated with AlphaFold3.

Results. NCAM2 knockdown perturbed cell adhesion and gastrulation, resulting in embryos arresting at the blastula stage and cells protruding from the epithelium. It also caused the loss of basal localization of Cadherin-3. AlphaFold3 structural predictions suggested that NCAM2 forms a strong dimer between immunoglobulin and fibronectin domains of opposing proteins, consistent with a role in adhesion. HCR in situ hybridization of *snailA* and *brachyury* revealed that NCAM2 and -catenin knockdown embryos exhibited similar phenotypes, including reduced *brachyury* expression at the blastopore, expanded *snailA* expression in the mesoderm, loss of body axis patterning, and failed gastrulation. Treatment of NCAM2 and -catenin knockdown embryos with azakenpauellone, a selective inhibitor of GSK3 in the -catenin destruction complex, rescued gastrulation and restored *snailA* and *brachyury* expression to near-baseline levels.

Conclusion. NCAM2 is required for epithelial integrity, cell adhesion, gastrulation, and axial body patterning in *Nematostella* embryos. Since Cadherin-3 remains highly expressed in NCAM2 knockdown embryos, the loss of adhesion alone does not explain the cell protrusions. NCAM2 knockdown also disrupts body axis patterning similar to -catenin knockdown, with expansion of *snailA* and reduction of *brachyury* expression. Both phenotypes responded to azakenpauellone treatment. This suggests that NCAM2 may regulate signaling pathways that regulate cell migration and body axis patterning.

Biology

Analyzing the functional role of Adad1

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Purpose. There is an incomplete understanding of how genetic factors lead to infertility. We use zebrafish as a model organism to learn more about these factors. Adad1 (Adenosine deaminase domain-containing protein 1) is a gene that has been shown to affect male fertility, and previous research in our lab has found that Adad1 is essential for germ cell maintenance in zebrafish, with a null mutant leading to total loss of germ cells. This gene encodes a double-stranded RNA-binding motif (dsRBM) and an adenosine deaminase domain (DD). We hypothesize that the dsRBM and DD domain each play a significant role in the functional activity of Adad1. Ultimately, we hope to gain insight into RNA-binding proteins' role in germ cell maintenance in both zebrafish and humans.

Methods. We performed an in vitro assay in zebrafish embryos, looking at the RNA regulation by Adad1 and the effects of individual domain knockouts on protein function. In this tethering assay, we tethered our protein of interest to MS2, a bacteriophage coat protein that binds to RNA stem loop helices. When mixed with firefly Luciferase mRNA with these motifs, it leads to a measurable luminescence output after allowing translation to occur. We synthesized the RNA and injected it into zebrafish embryos at the one-cell stage. The tethering assay was repeated with Adad1 domain knockouts.

Results. Our initial tethering assay revealed that Adad1 had a significantly higher luciferase protein produced than our control ($p < 0.05$). Based on our hypothesis, we expected that if one of the domains were necessary for Adad1's functional activity, it would produce lower luciferase proteins. The tethering assay with the Adad1 knockouts significantly differed from MS2, our baseline luminescence, except for the effector protein that only had Adad1 DD.

Conclusion. Neither domain seems necessary for the effect Adad1 has on proteins. Our preliminary results indicate that Adad1 interacts with RNA to increase protein production, but future experiments are needed to determine the exact mechanism.

Extremophilic Microbial Isolates From Alaskan Iron Seep Demonstrate Diverse Characteristics

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Purpose. Submitted via email to Madi

Methods. Submitted via email to Madi

Results. Submitted via email to Madi

Conclusion. Submitted via email to Madi

Quantifying Stress-Induced Genomic Variation from Sequencing Data

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Purpose. Certain species of bacteria in the gut microbiome cause disease, but treatment with antibiotics indiscriminately kills harmful and beneficial species. Bacteriophages, viruses that infect specific bacterial species, offer a more targeted treatment that could eliminate harmful species while maintaining beneficial gut microbes. While investigating the ability of the bacteriophage KL11 to kill the pathogenic gut bacterium *Escherichia lenta*, we found that KL11 kills *E. lenta* in vitro but not in vivo (mouse gut microbiome). We aimed to test whether KL11 treatment selects for structural variants (SVs) of *E. lenta* that may confer resistance.

Methods. We colonized 8 germ-free mice with *E. lenta* and infected them with either active KL11 (n=4) or heat-killed KL11 (n=4). After 4 days, we collected stool from each sample to sequence the DNA of the *E. lenta* in the gut microbiome. We used the computational tool Flye to create de novo genome assemblies from long-read sequencing data and performed whole-genome comparison using Mummer. Next, we created a pipeline to quantify the proportion of *E. lenta* strains in each sample using the tools Minimap, Samtools, and Sniffles. We also validated our pipeline predictions using simulated long reads.

Results. All genome assemblies from active-phage samples contained a large 32-kbp inversion (SV1) absent in the assemblies from heat-killed samples. We labeled this inversion SV1 and quantified its abundance across treatment groups. SV1 comprised an average of 4.1% of the *E. lenta* population in heat-killed controls versus 43.8% of the population in the active-phage group ($p = 0.002$). Our predicted strain abundance for our samples correlated with qPCR estimates ($p = 0.007$), an independent method for measuring structural variant proportions. We validated our pipeline using synthetic long reads simulated from reference genomes, and the predicted variant abundances closely matched the simulated proportions ($p < 0.001$).

Conclusion. In conclusion, this investigation demonstrates that treatment of *E. lenta* with active KL11 phage selects for SV1, indicating SV1 confers resistance. Understanding the mechanism of resistance in SV1 strains will help researchers design effective bacteriophage interventions for harmful gut bacteria. This research also shows that long-read sequencing can be used to quantify structural variants from in vivo samples.

Quantifying biofilm exopolysaccharide expression in *Pseudomonas aeruginosa* clinical strain CF23

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Purpose. To investigate the biofilm forming capacity of the *P. aeruginosa* CF23 strain by quantifying the expression of biofilm associated genes and then correlating gene expression data with phenotypic traits such as motility and EPS production assays in order to understand the underlying molecular mechanisms that contribute to the strain's chronic infection phenotype.

Methods. For phenotypic analysis, colony morphology and EPS production were measured using Congo red plates. Furthermore, motility assays were conducted. Twitching motility was assessed by staining with Coomassie Blue to visualize twitching zones at the agar–petri dish interface. Swimming motility was measured by diameter of migration on LB plates containing 0.3% agar. For gene expression analysis, RNA was reverse-transcribed into cDNA to perform quantitative PCR with primers targeting *pelA*, *pslD*, *algD*, and the housekeeping gene *rplU*. Amplification was carried out, and product specificity was confirmed by melt curve analysis. Amplification products were further validated by gel electrophoresis.

Results. CF23 (clinical strain) and PA15 (Pel constitutive mutant) displayed stronger Congo red binding when compared to Pel deficient strains (PA25 and PA26). Pel-deficient strains (PA25, PA26) exhibited larger twitching and swimming zones compared to high-EPS strains (CF23, PA15), suggesting that reduced EPS production enhances Type IV pili- and flagella-mediated motility. CF23 expressed *pelA*, *pslD*, and *algD*, confirming robust production of all three EPS types at the transcriptional level.

Conclusion. These findings suggest that *Pseudomonas aeruginosa* CF23 strain possesses strong biofilm forming capacity, driven elevated EPS production of Pel, Psl and alginate. The correlation between gene expression data and phenotypic traits such as decreased motility but strong biofilm may suggest how the strain has adapted to chronic infection conditions.

ROR1 CAR T-Cells As a Novel Treatment for Prostate Cancer

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Purpose. Chimeric Antigen Receptor (CAR) T-cells engineered to kill tumor cells have revolutionized the treatment of hematologic cancers. To do this, CAR T-cells use an artificially expressed antibody to bind to surface antigen on tumors. The efficacy of this approach against solid tumors has been limited, which is primarily due to (1) the absence of appropriate targets from solid tumors and (2) the presence of immunosuppressive factors like TGF in the tumor microenvironment. Our lab is for the first time investigating the tumor-targeting ability of novel CAR T-cells specific for the Receptor Tyrosine Kinase-like Orphan Receptor 1 (ROR1) antigen.

Methods. We analyzed ROR1 surface expression on various prostate cancer cell lines by flow cytometry. Next, tumor cells underwent GFP-luc lentivirus transduction and were sorted to yield pure target cells for cytotoxicity assays. Finally, we evaluated the tumor-killing ability of ROR1-specific CAR T-cells and compared them to ROR1 CAR T-cells armored with a dominant-negative form of TGFRII.

Results. Our novel CAR T-cells were able to bind to and produce various cytokines in response to recombinant ROR1 protein. We found that most of the prostate tumor cell lines express substantial ROR1 levels. Finally, unmodified and “armored” CAR T-cells both efficiently killed prostate cancer cells, even at low effector/target ratios.

Conclusion. Our findings suggest for the first time that ROR1 CAR T-cells may serve as a promising treatment for prostate cancer, especially when combined with anti-TGF mechanisms. Additional studies are underway in our lab evaluating this novel CAR T-cell approach in other solid tumors.

Using Calcium Imaging to Investigate How Flow Direction Influences Habituation in Zebrafish Lateral Line Hair Cells

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Purpose. This study investigates how the direction of water flow influences habituation in oppositely oriented hair cells in the lateral line of larval zebrafish by pairing genetically encoded calcium indicators with fluorescent microscopy to visualize calcium transients in hair cells. Examining directional selectivity in habituation can provide insights into zebrafish behavior, from predator avoidance to schooling, and further our understanding of how sensory systems adapt to repeated stimuli.

Methods. This study uses 4–7 days post-fertilization zebrafish larvae with MyoGCaMP8m, a genetically encoded calcium indicator, encoded in their hair cells. Prior to measuring calcium signals, the larvae are submerged in MS-222, a temporary anesthetic. The larvae are then immobilized with tungsten pins and injected with -bungarotoxin to prevent motion artifacts during imaging. After immobilizing the larvae, calcium transients were visualized in lateral-line hair cells in response to fluid jet stimulation. Hair cells were stimulated with the fluid jet in both anterior-to-posterior and posterior-to-anterior directions, and fluorescence was imaged using fluorescent microscopy. Imaging data were acquired with Micro-Manager and Igor software, and analyzed using Fiji to quantify fluorescence over time.

Results. The results of this study indicate that hair cells oriented in the A-to-P direction displayed lower peak fluorescence and a faster decline in calcium signal over time compared to P-to-A oriented cells. Further, when comparing peak fluorescence relative to the previous baseline, posterior-facing hair cells exhibited faster rates of decreased fluorescence, suggesting greater habituation compared to anterior-facing hair cells. These findings suggest orientation-specific differences in repetitive stimulation between posterior and anterior-facing hair cells.

Conclusion. This study provides preliminary evidence that hair cell orientation in zebrafish lateral-line neuromasts influences habituation. The different calcium responses between A-to-P and P-to-A hair cells suggest that orientation plays a role in how zebrafish respond to repetitive stimuli. These findings support further investigation into how sensory systems adapt to stimulus direction and frequency, how calcium signals differ with agonists, and whether the habituation asymmetry exists at the afferent neuron level.

Chemistry

Crystallization of Catalase from Bovine Liver

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Purpose. This study attempts to replicate and modify a relatively old method of crystallizing catalase from solid liver. The method that inspired the experiment is from the 1950s and provides a number of ways in which enzyme can be crystallized depending on the type and amount of liver.

Methods. Solid liver was crushed, filtered, and decanted to isolate the enzyme. Two livers were used following the same procedure, although the second was carried out much more efficiently. Acetone was frequently added to the solution in order to precipitate the enzyme. After two rounds of vacuum and gravity filtration, the resulting solution was allowed to dialyze for several days before the enzyme was expected to crystallize.

Results. No crystals were detected upon completing our experiment. First, we expected a precipitate to form after several days of dialysis, but this was not the case. There are several reasons that this is possible, with the foremost being that our earlier rounds of filtration were likely not effective enough at removing lipids and other large molecules. Failing to obtain this precipitate, we extracted the solution from the dialysis tubes and attempted to “recrystallize,” a method that would have purified any precipitate that we had collected. This effort also failed to yield crystals. Dr. Kendra Marcus used a microscope to help our search but we were unable to find any crystals.

Conclusion. Our experiment failed to isolate and crystallize catalase enzymes after 48 hours, likely due to shortcomings in the early filtration steps. In the future, we may experience better results with more selective filtration as well as by allowing more time for dialysis and recrystallization.

Designing a high throughput screen for activators and inhibitors of the SH2-domain containing protein tyrosine phosphatases 1 (SHP1)

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Purpose. SHP1 plays a role as both a tumor suppressor and promoter, so exploring its activity can clarify SHP1's role in cancer development. Small molecules can be used to potentially shift the conformational equilibrium of SHP1 between the active and inactive states. However, no well-characterized allosteric activators or inhibitors currently exist. This project focuses on a mutant (S496P) that is positioned halfway between the closed and open states, exhibiting increased activity, as a tool for screening small-molecule candidates.

Methods. Wild-type SHP1 and truncated SHP1 proteins were expressed and purified. and purity was confirmed by SDS-PAGE. Enzyme kinetics were measured using a Michaelis–Menten assay to determine catalytic parameters. Inhibition assays were performed by testing five small-molecule compounds across varying concentrations to assess their effects on SHP1 activity.

Results. Kinetic analysis revealed that full-length WT SHP1 displayed more than a threefold lower catalytic efficiency compared to truncated SHP1, highlighting the importance of the C-terminal tail. Inhibition assays produced ambiguous results: while some compounds showed reduced activity at higher concentrations, others unexpectedly increased SHP1 activity.

Conclusion. Although we were able to successfully express WT SHP1, the results from the inhibition assay were ambiguous, with no consistent trend observed across the compounds. Therefore, the assay will be repeated to confirm whether these molecules truly inhibit SHP1 activity. We also plan on expanding our screening to a broader library of activators and inhibitors. A particular focus will be placed on evaluating the effects on the S496P SHP1 mutant.

Developing an Optical Microscopy Method to Detect Microplastics in Environmental Samples

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Purpose. Microplastic contamination in coastal environments poses a growing threat to ecosystems and human health, yet current detection methods for these microscopic particles are often limited by long processing times, high costs, limited availability, and low signal detection for complex samples. This project aims to develop an optical microscopy technique using Differential Dynamic Microscopy (DDM) to identify and characterize microplastics in environmental water samples. DDM analyzes light intensity fluctuations to extract information about particle dynamics and properties. This approach could enable effective characterization and differentiation of microplastic types, enhancing tracking, monitoring, and mitigation efforts.

Methods. Microplastic samples composed of polystyrene and polymethyl methacrylate (PMMA) were suspended in water and imaged using brightfield microscopy. Videos of particles were recorded at varying focal planes to capture particle dynamics across sample depth. These videos were processed using a DDM code to extract quantitative parameters, including amplitude as a function of the wavevector (q) and diffusivity. Additional image analysis was conducted to examine and quantify ring patterns produced by particle light scattering at varying axial positions.

Results. Analysis revealed that diffusivity increased with height from the substrate bottom, eventually reaching a plateau near the middle of the sample—consistent with expected Brownian motion behavior. The amplitude function, $A(q)$, exhibited local minima corresponding to the particles' effective length scale. When plotted against axial position (z), the amplitude peak followed a Gaussian distribution. Fitting this profile enabled the determination of the particle's refractive index, yielding a value (~ 1.53) that aligns closely with the known range for polystyrene (1.55–1.59).

Conclusion. These findings lay the foundation for an efficient, accessible, and label-free method of microplastic identification and provide insight into how DDM can be adapted for complex, real-world samples. Future work will focus on quantifying and refining measurement error, as well as expanding the analysis to include additional polymer types and mixed samples to broaden applicability and validate trends.

Electrodeposited Bimetallic Catalysts for Impurity-Tolerant Electrocatalytic Hydrogenation of Muconic Acid to Adipic Acid

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Purpose. The application of electrocatalysis to biomass conversion, biobased materials, and bioenergy is a promising approach for transitioning away from a fossil fuel-dependent economy. Of particular interest is the synthesis of adipic acid, an industrially valuable compound and a key monomer of nylon 6,6. This study examines the electrocatalytic hydrogenation (ECH) of fermentation-derived *cis,cis*-muconic acid (ccMA) to adipic acid (AA), with an emphasis on tuning the catalyst composition to enable efficient conversion directly in unpurified fermentation broth. This integrated approach has the potential to improve overall process efficiency by eliminating the need for downstream purification steps to extract ccMA before electrocatalytic conversion. A key challenge in this approach is catalyst deactivation by biomolecules, particularly sulfur-containing amino acids, that are inherently present in the fermentation broth. Previous work in our group has shown that poisoned catalysts can be regenerated in the absence of these impurities using electrochemical techniques like cyclic voltammetry (CV) and reductive stripping. Therefore, developing robust and impurity-tolerant catalysts is crucial for enabling the direct use of fermentation broth as the reaction medium.

Methods. One potential strategy to improve sulfur tolerance is by modifying the electronic structure of the catalyst through alloying. Electrodeposition provides an effective and low-cost method for synthesizing such alloyed catalysts, offering uniform deposition and simple implementation. In this study, PdAu, PdRu, PdBi, and PdPt bimetallic catalysts were deposited on porous nickel foam using chronoamperometry (CA). The resulting catalysts were tested for the electrocatalytic hydrogenation of ccMA to AA, and quantitative ¹H-NMR spectroscopy was used to characterize the production of adipic acid.

Results. The PdAu/Ni catalyst outperformed all other catalysts, reaching near full conversion of ccMA after 3 hours. Scanning electron microscopy (SEM) images of the catalysts showed a wide distribution in size and morphology of electrodeposited metals, while energy dispersive X-ray spectroscopy (EDS) and X-ray diffraction (XRD) were performed on the PdAu/Ni catalyst to confirm deposition of palladium and gold over the nickel foam surface and alloying of palladium and gold.

Conclusion. A set of metal and metal alloy electrocatalysts synthesized through electrodeposition were screened to investigate their effectivity in converting ccMA to AA. Future directions include testing catalysts for poisoning resistance. Investigation of the electronic properties of alloys through density functional theory (DFT) and other computational methods may suggest other bimetallic catalysts to be experimentally tested in the lab.

Estimating Enthalpies of Solution Using Infrared Thermography and Coffee Cup Calorimetry

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Purpose. To create a general chemistry lab featuring infrared thermography, we used the FLIR ONE camera to estimate the enthalpy of solution of the following ionic solids: sodium chloride, sodium bromide, lithium chloride, lithium bromide, potassium chloride, potassium bromide, and ammonium chloride.

Methods. By measuring the difference in temperature before and after the solute was dissolved, we determined if the reaction was endothermic or exothermic. Furthermore, we used the formula $-H_{\text{sol}} = q_{\text{H}_2\text{O}} = mcT$ (with C_{cal} accounted for) to estimate the enthalpy of solution, which we subsequently compared to the literature values.

Results. We found potassium chloride, potassium bromide, and ammonium chloride to be significantly endothermic; sodium chloride and sodium bromide to be near zero; and lithium chloride and lithium bromide to be significantly exothermic. The observed trend likely occurred because ions of smaller atomic radii have stronger interactions with water, resulting in greater ion-dipole interactions and more exothermic enthalpies of solution. All such values roughly correspond to the literature values, though the magnitude of their estimated enthalpy of solution was reduced by about three times. For more accurate measurements in the future, the specific heat capacity of the solutions can be found using a differential scanning calorimeter (DSC).

Conclusion. Overall, merging coffee cup calorimetry with infrared thermography offers an effective way to estimate enthalpies of solution and visualize heat transfer. We conclude that this experiment would successfully educate general chemistry students on thermodynamics-related concepts.

Investigating Electronic and Steric Modulation in Group V Supported Titanium Alkoxides for Controlled Ring-Opening Polymerization

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Purpose. To determine how electronic donor strength and steric bulk of (Group IV) donor scaffolds on Ti(IV) alkoxide initiators govern initiation/propagation kinetics and polymerization control in the ring-opening polymerization of cyclic esters, so we can establish design rules for living, predictable ROP that yields biodegradable polymers.

Methods. Ligands were assembled from commercially available precursors using literature-adapted routes and prior in-lab protocols. Structures (and purity) were confirmed by NMR (¹H; others as applicable) and by SCXRD when suitable crystals were obtained. Air-stable ligand steps were performed in a fume hood; air-sensitive Ti (IV) alkoxide complexes were synthesized and handled exclusively in an anhydrous, O-free glovebox. Ring-opening polymerizations of cyclic esters were run neat (melt conditions) by combining monomer and catalyst at 140 °C inside the glovebox.

Results.

We crystallized several titanium complexes and found a binding geometry that differs from prior reports. Their structures were confirmed by SCXRD and supported by solution NMR. In melt ROP, a representative Ti catalyst achieved 97% monomer conversion in 1h and produced a polymer with a DP of 75. We will use size-exclusion chromatography to measure molecular weights and dispersity, assessing polymer quality. Overall, these results indicate a promising Ti platform, allowing us to compare the effects of different substituents on structure and performance.

Conclusion.

We examined the kinetics and performance of titanium alkoxide catalysts for ring-opening polymerization of cyclic esters. By varying ligand functional groups, we directly compared how changes in donor electronics and steric bulk affect catalyst activity and control. The modular ligand set provides many new derivatives to explore next. Overall, the data support continued development of Ti-based systems to produce biodegradable polymers with targeted properties and predictable control.

Towards the Synthesis of Borangulene for Cocrystalization with TOPA

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Purpose. We attempted to synthesize borangulene, a planar, boron centered analog of the bowl-shaped, nitrogen centered molecule TOPA. TOPA can interconvert between two stable forms—bowl up and bowl down—and may be used for binary data storage on the molecular scale. However, further computational analysis revealed that the flat form also corresponds to a potential well, i.e., the molecule has three stable forms. Polymerizing TOPA with borangulene through a Lewis pair bond between the nitrogen and boron centers may force the bowl up and bowl down geometries to be much more favorable, thereby shifting the potential energy curve of TOPA into a double well.

Methods. Referencing the first total synthesis of borangulene in literature(1), we attempted a modified route. The first step combines 2-bromoresorcinol and 2-bromo-1-iodo-3-methoxybenzene in a 1:3 molar ratio in an Ullmann-type reaction. After the reagents were added to a dry 3-neck round bottom flask under nitrogen gas, the mixture was stirred for 20 hours at 110°C. The mixture was passed through a celite plug using chloroform, washed with saturated NH₄Cl and brine, then separated via column chromatography with 95:5 hexanes:ethyl acetate as the eluent. A sample was taken after the liquid-liquid extractions and analyzed using gas chromatography-mass spectrometry (GCMS), along with the two starting materials.

Results. The Ullmann-type reaction seemed to have a poor yield, and the product was difficult to separate via column chromatography. After several unsuccessful attempts, an alternative route which takes several steps from the synthesis of TOPA was explored. The GCMS chromatogram for the reaction mixture showed an intense peak matching 2-bromo-1-iodo-3-methoxybenzene, and no significant peak matching 2-bromoresorcinol. The chromatogram for 2-bromoresorcinol showed several peaks.

Conclusion. Chromatograms revealed that the limiting reagent (2-bromoresorcinol) was consumed, and that the 2-bromoresorcinol sample was quite impure, the intermediate tri-brominated resorcinol being the major species. However, the synthesis of borangulene with an Ullmann-type reaction may still be viable if 2-bromoresorcinol can be purified further. The alternative route which follows the synthesis of TOPA in its first steps should still be explored.

(1) Yuichi Kitamoto; Suzuki, T.; Miyata, Y.; Kita, H.; Funaki, K.; Oi, S. The First Synthesis and X-Ray Crystallographic Analysis of an Oxygen-Bridged Planarized Triphenylborane. *Chemical Communications* 2016, 52 (44), 7098–7101. <https://doi.org/10.1039/c6cc02440h>.

Computer Science

Don't Be Rude, Daisy: Socially-Aware Robot Navigation

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Purpose. This project investigates how Boston Dynamics' quadruped robot Spot, nicknamed Daisy, can be equipped with socially-aware navigation skills in human-populated environments. The goal is to enable Daisy to move respectfully through dynamic spaces by modeling behaviors that prevent crowding, support group integration, and adhere to human proxemics.

Methods. A real-time multi-person detection and tracking pipeline using YOLOv7 and StrongSORT was implemented to allow Daisy to perceive and identify individuals across frames. Full 360-degree perception was achieved by streaming all five onboard cameras in parallel using multithreading. Depending on social context, Daisy uses either the Social Forces Model (for individual interactions) or the BOIDs model (for group movement). The Social Forces equation uses dispersion constants to control attraction to goals, repulsion from humans, and avoidance of obstacles. The BOIDs model includes weights for separation, alignment, and cohesion, guiding Daisy to move within a group as an active participant rather than trailing behind.

Results. Daisy is able to detect and consistently track people across multiple viewpoints and estimate their relative velocities and positions. The use of both SFM and BOIDs allows flexible navigation strategies depending on social context. Initial field tests show stable multi-camera tracking, successful person-following behavior, and consistent avoidance of personal space violations.

Conclusion. This study demonstrates the feasibility of applying hybrid navigation models to enable context-aware, socially compliant movement in a mobile quadruped robot. These behaviors were driven by manually chosen parameters, but future work will focus on optimizing these parameters through reinforcement learning in a simulated environment to improve adaptability and realism in navigation behavior.

Effects of Reassessment Policy on Emergency Department Performance

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Purpose. Prolonged length of stays (LOS) in emergency departments results in adverse long and short-term effects for high acuity patients, increasing death and "left without being seen" (LWBS) rates. Misdiagnosis, intensified by staff fatigue and patient congestion, is a key factor in high LOS, especially when uncorrected. A proposed solution to reduce LOS and correct misdiagnosis is to reassess patients after post-triage delay.

Methods. We developed a discrete-event driven simulation of the emergency department (Sort → Registration → Triage → Treatment Zones) with dynamic patient arrivals, variable staffing, and probabilistic misdiagnosis to evaluate reassessment strategies. Three models were tested: (1) baseline with no misdiagnosis or reassessment, (2) misdiagnosis only with increasing error variability and rate, and (3) misdiagnosis with post-triage reassessment at delays from 0 to 240 minutes. Each model was simulated for one year and repeated ten times to measure LOS, LWBS, and deaths.

Results. Reassessment after 120 minutes optimizes outcomes. When misdiagnosis variability and rate are low to moderate during this delay, deaths and LWBS rates decrease significantly without extending LOS because more high acuity patients in congested zones are corrected. Any shorter delay reduces reassessment efficiency, while longer delays allow more untreated high-acuity patients to deteriorate, increasing LOS due to treatment complications. LWBS rates were less sensitive to reassessment timing.

Conclusion. Reassessment after 120 minutes in post-triage delay is effective. Future work will compare alternative staffing models for reassessment, such as dedicating staff for reassessment versus rerouting patients back to triage.

Effects of Swing Team Routing Policy on Emergency Department Performance

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Purpose. Emergency Department (ED) crowding remains a critical issue, with prolonged length of stay (LOS) negatively impacting patient outcomes. This study applies queueing theory to evaluate policies for routing patients to a flexible 'Swing team' of providers, with the objective of reducing LOS for moderate-acuity patients.

Methods. In this study, we use a simulation modeled on patient flow at Mt. Sinai Hospital ED. We tested four distinct policies to assign patients from triage to the treatment zone. These included Join-the-Shortest-Queue (JSQ), which routes patients to any zone with the shortest line, and Swing Extends Red (SER), where the Swing team assists the high-moderate acuity 'Red zone'. We also tested two load-balancing policies: Idealized Load Balancing (ILB), which uses staffing levels and queue lengths to route patients to the zone with the shortest expected wait time, and Constrained Load Balancing (CLB), which first directs all high-moderate acuity patients to the Red zone before applying the ILB logic for other patients.

Results. Results indicate that ILB achieves the lowest LOS overall (a 58.9% reduction from baseline with no Swing team) but violates operational constraints by routing high-moderate acuity patients to zones other than Red. In contrast, CLB balances workload while adhering to the zone constraints, reducing LOS by a substantial 31.5%. SER performs best when swing staffing is low, while ILB and CLB excel under higher swing capacity.

Conclusion. In conclusion, workload balancing policies show significant promise for improving ED efficiency. The CLB policy, in particular, emerges as the optimal and implementable strategy. It effectively balances the high performance of an idealized model with the practical, real-world constraints of patient care, significantly reducing LOS. These findings demonstrate that routing policies are a powerful tool to effectively utilize the Swing team's potential to contribute to ED performance.

Encoding, Splitting, and Solving: Tackling Chvátal's Conjecture with SAT

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Purpose. This study addresses Chvátal's Conjecture, a long-standing problem in extremal combinatorics, which states that in any downset, one of the largest intersecting families is a star. An intersecting family is a set of subsets where the intersection of all pairwise sets is nonempty, and a star is an intersecting family where there exists a common element contained in all sets. Previous research safely verified the conjecture up to $n = 7$ (using an IP solver).

Methods. The existing IP formulae were encoded into conjunctive normal form, and then solved using Kissat, a state-of-the-art SAT solver. Several model reductions were applied: (1) assertion of generalization of sets by grouping sets by their symmetric differences, (2) splitting cardinality constraints into cases and (3) splitting on the maximum and minimum size of subsets in the intersecting family in order to tackle each subproblem individually. For additional speedup, Cube and Conquer was used on the hardest subproblems. All problems were run on FrostByte, Amherst College's high-performing computing cluster.

Results. For $n = 7$, our model reductions achieved a 4.15x speedup over the IP solver and a 70.09x speedup over the original SAT encoding. The total runtime was 41.9 seconds. We were able to safely solve $n = 8$ for the first time, with a total runtime of 422578.40 seconds. For the hardest subproblem, Cube and Conquer achieved an additional 3.84x speedup (reducing runtime to 4067 seconds).

Conclusion. By encoding the existing IP formulae into SAT and splitting cardinality constraints and the minimum and maximum sets in the intersecting family, a significant speedup was achieved for $n = 7$ and, moreover, $n = 8$ was safely solved for the first time. We also showed that the hardest subproblems can be further partitioned into cubes and then solved in parallel, which allowed for further performance gain.

Nonlinear Independent Component Analysis can identify spatial structure from the neural activity of hippocampal CA1 neurons

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Purpose. The purpose of this study is to investigate whether spatial structures can be extracted from high-dimensional neural recordings. Previous work has shown that populations of neurons, including place cells and boundary vector cells, encode spatial information. However, neural activity is often high-dimensional and nonlinear, making it unclear whether and how spatial structure can emerge in low-dimensional representations. To extract latent structures from high-dimensional neural activity, we applied a nonlinear Independent Component Analysis (ICA) approach to neural data collected during spatial navigation tasks. In particular, we adopted CEBRA (Consistent EmBeddings of high-dimensional Recordings using Auxiliary variables), a recent contrastive learning framework that leverages auxiliary variables.

Methods. The experiment consisted of two phases. In the preliminary phase, we used a synthetic neural dataset to introduce complementary features and enhance data diversity. In the main phase, we analysed hippocampal Cornu Ammonis area 1 (CA1) recordings from freely moving rats to examine whether the insights gained from the synthetic data generalize to real neural dynamics. CEBRA was applied in both discovery-driven (self-supervised) and hypothesis-driven (supervised) modes to extract low-dimensional, behaviourally relevant latent representations from the neural activity.

Results. In the preliminary experiments, the discovery-driven mode produced latent representations whose structures closely matched the geometry of the corresponding environments. To further assess structural alignment, we trained embeddings in the hypothesis-driven mode using the rats' positional coordinates as auxiliary variables, and quantified the similarity between embedding and position using the R^2 score from a linear regression fit. We applied the same procedure in the main experiments. In the preliminary experiments, the discovery-driven mode achieved a higher R^2 score (~ 0.94) than the hypothesis-driven mode, indicating that spatial structure could be directly extracted from the autocorrelation of neural data. In contrast, in the main experiments, the hypothesis-driven mode achieved superior performance (~ 0.91), suggesting that conditioned non-stationarity in neural activity can more effectively reveal spatially relevant structure.

Conclusion. Our findings reveal that rich spatial features are embedded in neural activity and can be recovered through suitable low-dimensional embeddings. The spatial structure observed in the learned latent space suggests that neural population activity may be organized along a low-dimensional manifold reflecting the geometry of physical space.

Geology

AI Going Against the Grain: Testing a Neural Network's Ability to Identify Microtextures in Quartz Sand

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Purpose. As grains of sand are transported across Earth's surface, they acquire distinctive surface textures characteristic of their environment. Rivers, glaciers, waves, and wind all produce different textures. The distribution of microtextures on individual grains can be used to reconstruct the sedimentary history of a sediment. SAND AI is a Deep Neural Network (developed by Hasson et al. 2024) intended to address the limitations of microtextural analysis by providing an objective and qualitative method of determining depositional environments. This study investigated the extent to which successive sedimentary cycles can be derived from the model's output, using samples with known sedimentary histories.

Methods. The model was developed and trained (by Hasson et al. 2024) to distinguish between glacial, eolian, fluvial, and beach transport, using sand grains from known environments. It then underwent external validation on sediment from modern sites unknown to it. We used SAND AI to analyze sediments from environments it was trained to recognize, in addition to those it was not. Each sample was cleaned in hydrogen peroxide and boiling HCl in order to remove surface coatings. Grains were imaged both before and after cleaning in order to gauge the effectiveness of the cleaning procedure. Quartz grains were randomly selected under a microscope and imaged on a Scanning Electron Microscope (SEM). A spectrum was taken on a clean face of every grain analyzed to verify that it was quartz. Images were then processed by the SAND AI model.

Results. The model proved better at accurately identifying glacial and eolian environments than beach and fluvial ones, and we found that glacial microtextures linger on grains that are subsequently transported in other environments. Samples for which fewer than 5 grains were identified at $\geq 75\%$ confidence are considered to have not been successfully identified by the model. The one lithified sample tested was incorrectly identified, which may be due to diagenetic alteration of microtextures and cement growth.

Conclusion. Though we conclude that SAND AI can determine a sediment's past environments, with the current model output we cannot confidently form a chronological sedimentary history without further observation, because it does not recognize cross-cutting relationships between microtextures. In addition, more work is needed to determine if the model can successfully be applied to uncover the transport history of lithified sedimentary rocks, or if diagenetic alteration renders the model inaccurate.

From Data to DEM: Generating Improved Basal Topographies of Petermann Glacier Using Novel Swath Radar Data Processing

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Purpose. Outlet glaciers are key sites of mass loss from the Greenland Ice Sheet, and projections of future mass loss are sensitive to the prescribed bed in ice sheet models. Increased resolution of the basal topography results in more accurate predictions, but measurements from flights over the ice sheet are limited in resolution and coverage. Generating swaths from cross-track radar data provides additional coverage by distinguishing the direction of the received radar signals. We can use these swaths to produce improved Digital Elevation Models (DEMs) of basal topography through kriging interpolation.

Methods. We applied swath techniques to select radar data collected over a 32,400 km² area near Petermann Glacier in Greenland, and used a bed tracking algorithm to identify which radar signals corresponded to reflections off of the bed of the ice sheet, generating the bed picks for the swaths. The variations in data quality and errors in the bed tracking algorithm required the use of a flagging and retracking algorithm to improve measurement accuracy, which removed and regenerated bed picks based on the signal strength at the originally identified bed pick. The swath radar data is converted to xyz points, calculated based on the speed of light in ice and air, the direction of arrival, and the two way travel time of the received signal. After removing outliers and performing a crossover analysis of overlapping measurements, the data was input into a kriging algorithm that performed a geostatistical interpolation of the area in between the existing radar data.

Results. The data processing algorithms reduce crossover errors where two swaths measure the same location, and indicate that some swaths (such as from the 2010 season) are unsuitable as kriging input due to poor resolution of the cross-track data. The resulting interpolation from combined swath and nadir data using the GStatSim Python package reveals artifacts from the input data in both, but adding the swath data resulted in increased texture in the areas where they were present, though it's uncertain how much is due to the artifacting from the swaths themselves.

Conclusion. The DEM outputs from this study will require future processing to remove the artifacts from the input data before any quantitative comparisons can be made, but future DEM generation can also incorporate swath data in other ways which were not tested here, such as informing kriging parameters like local anisotropy. Improving swath generation and kriging techniques has the potential to generate higher resolution DEMs, and improve our knowledge of glacier basal topography, for use in ice sheet modeling and glacier geomorphology analyses.

Is Nye Theory Accurate or Antiquated: Verifying Modeled Crevasse Depths Against Radar-Derived Crevasse Depths in Pine Island and Thwaites Glaciers

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Purpose. There has been no comprehensive testing of the accuracy of Nye's theory of crevassing in predicting crevasse formation and depth. This study models basal crevasse depths according to the principles of Nye and corroborates them against manually derived crevasse depths from radar data in the high strain-rate Pine Island and Thwaites glaciers.

Methods. Using the Nye crevasse depth formula and using skin temperature, ice thickness, and velocity datasets for our calculations yielded our map for expected crevasse depths. Filtering through the available radar data for where the flight path is approximately parallel to ice flow direction, we collected crevasse tip and ice bottom depths, from which we determined the observed basal crevasse depths.

Results. We found that the true values of basal crevasse depth range from 1.64x to 2.52x the values predicted by our model. We found that less than 20% of the variability in observed values was explained by Nye's model of crevassing when using either the skin temperature or melting point as temperature and the model only accurately predicted 6% of basal crevasse depths.

Conclusion. This study provides strong evidence that Nye theory underestimates the depth of basal crevasses. A valuable question for further study is whether this underestimation could have been the result of incorrect values for the creep parameter A . In conclusion, it may be more appropriate to use newer models such as Linear Elastic Fracture Mechanics (LEFM) for use in predicting glacier system behavior.

Mathematics & Statistics

Is Lake Michigan a Metric (Dog-)Space?

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Purpose. In 2003, the mathematics community was introduced to a Welsh Corgi named Elvis in the paper "Do dogs know calculus?" In that article, author Timothy J. Pennings describes a game of fetch he played with his dog at Lake Michigan. With Elvis standing next to him at the shoreline, the author would throw a tennis ball into the water. The shortest route is a straight swim, while any route that involves running can be covered at a greater average speed than pure swimming. The problem for Elvis — and for generations of first-semester calculus students — is to find the precise path that minimizes the total travel time to the ball. Our research aimed to generalize the problem to include more paths.

Methods. As a problem rooted in distance, we sought to describe the problem in terms of a distance function, otherwise known as a metric. After describing a metric, we aimed to use that metric to numerically approximate the region Elvis could cover. The region that describes where Elvis can reach in x amount of time from point y is called the open ball of our metric beach-water space.

Results. The results of our research show that Elvis' path, be it water-to-water, land-to-water, or land-to-land, result in an extraordinarily shaped open ball. We learned that the shape was significantly more complicated than previously thought, requiring a non-elemental curve to describe part of the open ball.

Conclusion. In conclusion, our research investigated the optimal paths Elvis could take from various starting positions. Our work resulted in an open ball with an unconventional shape. Further steps would include determining the precise curve that outlines the open ball. Additionally, another point of investigation would be to create a more accurate metric that models the continuous velocity change at the shoreline.

Medicine (Basic & Translational)

Effect of Ethyl-3-Hydroxybutyrate on Viability of Pro-inflammatory Macrophages for Sepsis Treatment

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Purpose. Sepsis is an inflammatory condition, caused by aberrant host immune response to microbial infection, leading to increased tissue damage and mortality. The increased incidence of antibiotic resistance worldwide and lack of candidate drugs for sepsis treatment has increased mortality and morbidity in the USA & worldwide. Vivacelle Bio has been involved in developing phospholipid nanoparticles for treatment of sepsis, of which VBI-S has cleared Phase IIa clinical trials. Recently, we have also found a ketone body, ethyl-3-hydroxybutyrate (EHB), that can increase survival of severe sepsis in a cecal ligation and rat laceration model. In this model, CLL rats treated with 125mM EHB in aqueous phase at 8% of the rats body weight led to complete recovery within 36h. EHB (20mM) has been shown to demonstrate potent immunomodulatory effects, as it reduces pro-inflammatory cytokine/chemokine gene expression in M1 macrophages. However, it is unknown whether the reduced gene expression in M1 macrophages is linked to cell cytotoxicity by EHB.

Methods. THP-1 cells (50000) were seeded into 96 well plates and differentiated into M1 macrophages. Cells were treated in their respective groups for 24 hours. The post culture supernatant was removed, stained with trypan blue stain, and counted with a hemocytometer to determine the number of dead cells. The adhered cells were terminated with 4N HCl in isopropanol and the A570 was read. Statistical significance was determined by using Brown Forsythe and Welch ANOVA tests. Post Hoc analysis by Mann Whitney U tests were used to examine lower concentrations of EHB.

Results. It was found that EHB is cytotoxic at 67.5mM concentration, causing both decreased metabolic activity and an increased number of dead cells. However, EHB promotes metabolic activity at 10mM and 20mM concentrations, while also decreasing the number of dead cells, improving overall cell viability.

Conclusion. The decrease in pro-inflammatory cytokine/chemokine expression by EHB is caused by gene expression regulation at 10 and 20 mM instead of a cytotoxic effect lowering M1 macrophage populations. These findings suggest that mitigating the dysregulated inflammatory response with EHB could play a critical role in the effective treatment of sepsis. Future studies are necessary to develop a better understanding of the mechanism(s) by which EHB reduces inflammatory responses in M1 macrophages and enhances survival.

Neuroscience

Abundance of BK Channels in Mouse Inner Hair Cells Throughout the Onset of Hearing

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Purpose. Inner hair cells (IHCs) are the primary sensory receptors for hearing in the cochlea of mammals, and their expression of ion channels changes meticulously throughout hearing development. Among these channels, large-conductance calcium- and voltage-gated potassium (BK) channels are of particular interest as they generate the electrical currents that are necessary for mature hearing. This study aims to quantify the abundance of BK channels in mouse IHCs before, during, and after the onset of hearing in order to better understand their developmental role.

Methods. Because mice are born deaf and acquire hearing around postnatal day (P)10-12, three cohorts of mice (ages P8-P14) were euthanized for cochlear tissue collection. Western blotting was performed on nine samples to detect BK and glyceraldehyde 3-phosphate dehydrogenase (GAPDH) as a loading control. Band intensities were measured in ImageJ, corrected for background, and normalized to GAPDH. A standard curve was generated by loading increasing amounts of BK protein, confirming a linear relationship between band intensity and protein load. Antibody specificity was verified using BK knockout tissue.

Results. BK protein was detectable in three of nine samples, reflecting variability in tissue dissection and preparation. Expression was observed as early as P8, prior to hearing onset. In this sample, normalized BK intensity rose from 0.028 (P8) to 0.093 (P11; +232%). In two additional samples, BK levels increased from 0.067 (P10) to 0.212 (P14; +216%) and from 0.0295 (P10) to 0.0903 (P13; +206%). Due to small sample size and high variability between replicates, regression analysis was not feasible.

Conclusion. BK channels are detectable in mouse IHCs before the onset of hearing, suggesting a previously unrecognized role in development. Expression appears to increase with maturation, indicating potential involvement in the transition to functional hearing. These findings underscore the need for further study of ion channel regulation in IHC development, as such insights may inform gene therapy strategies aimed at regenerating hair cells and restoring hearing function.

Physics & Astronomy

A Model-independent Radio Telescope Dark Matter Search in the X Band

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Purpose. Despite mounting evidence for the existence of dark matter, its identification remains elusive. Recent detector technology advances have enabled experiments that are highly sensitive to specific dark matter candidates like the QCD axion. However, attention has recently been shifting to a theoretical framework for dark matter that does not assume specific models. Our astrophysical dark matter search therefore relies on as few assumptions as possible, focusing on the potential radiative decay or annihilation of ultralight dark matter in the Milky Way halo.

Methods. The X-band search utilized spectroscopic observations of 4999 Hipparcos catalogue targets from the Green Bank Telescope, collected between 2018 and 2025 as part of the Breakthrough Listen campaign. We applied a rolling normalization procedure to each spectrum to remove background variations while preserving potential dark matter signals. We then selected for spectral lines exhibiting the characteristic Doppler and intensity shifts expected from dark matter in the halo by constructing asymmetry spectra. To derive physics limits on the dark matter decay rate and velocity-weighted cross-section $\langle v \rangle$, we injected synthetic signals into raw spectra, calculated templates for specific dark matter physics parameters, and performed a p-value analysis on the cross-correlation between the asymmetry spectra and the templates.

Results. We fine-tuned the normalization scheme to maximize the signal-to-noise ratio of intensity asymmetries from synthetic dark matter signals. Optimal results were achieved with a rolling window size of $M = 7$ and a putative signal blanking region size of $m = 1.4$, where m represents the expected dark matter signal width. Our dark matter detection analysis in the 8000-8500 MHz frequency range excluded dark matter decay above 10^{-29} s^{-1} and annihilation above $\langle v \rangle 10^{-26} \text{ cm}^3 \text{ s}^{-1}$.

Conclusion. Our preliminary exclusion limits are consistent with results in the L and S bands (Keller et al. 2025), giving us confidence in the sensitivity of our X-band search. With our procedure optimized for the X-band dataset, we are now conducting the complete analysis across the C and X bands. In addition to confirming that our analysis is robust against uncertainties in solar velocity and dark matter virial velocity, we hope to evaluate how different dark matter halo models affect our exclusion limits.

Accretion in the Blue Ring Nebula's Stellar Merger Remnant: TYC 2597-735-1

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Purpose. Stellar mergers are a short-lived but common occurrence through the evolution of binary star systems. An opaque shell of gas and dust created during the merger obscures the remaining stellar merger remnant, making it nearly impossible to observe these objects. An ultraviolet blue ring nebula with a stellar merger remnant at its center was reported in 2020, making it the first merger system unobstructed by dust. Studying these remnants can provide new insights into the evolution of merger remnants, as well as the creation of peculiar stars like rapid rotators or blue stragglers, which are thought to come from stellar merger events.

Methods. This study sought to compare the observed spectrum of the stellar merger remnant to a model stellar atmosphere spectrum to assess the presence of excess ultraviolet continuum, a signature of active accretion occurring in a star. The observed spectrum used for this study was obtained from the Galaxy Evolution Explorer (GALEX) and spans 1300-3000 Å in the ultraviolet spectrum. The model used for comparison was selected by matching the metal abundance, surface gravity, and effective temperature parameters to the observed values. Regions with significant absorption features were masked in the observed and modeled spectra, allowing for a comparison of their continua. A chi-square goodness-of-fit test assessed whether the model spectra could predict the excess in ultraviolet emission in the observed spectrum. The goodness-of-fit test was performed for two regions: the entire observed spectrum with a binned far-ultraviolet region (1500-3000 Å) and the near-ultraviolet region only (2000-3000 Å).

Results. The goodness-of-fit test had a reduced chi-square value of ~25 for both regions it was performed on. These high values suggest that the model is not representing the observed data well.

Conclusion. Thus far, the high reduced chi-square indicates that the model poorly fits the data. These high values could be due to an excess in ultraviolet emission, but alternative explanations, such as underestimated uncertainties and model limitations, cannot be ruled out. Further research would help narrow down the source of the high reduced chi-square values and allow for a clearer interpretation of the data in the ultraviolet region. Additionally, higher-quality ultraviolet data and improved models will be beneficial in confirming whether the observed excess ultraviolet emission is evidence of active accretion in the merger remnant.

Atomic and Molecular Ion Trap

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Purpose. We trap and image an atomic ion crystal mixed with oxygen inside a linear Paul trap to take precision measurements of the vibrational frequency of oxygen. To improve the trapping process, we are switching the coolant from Beryllium ions to Calcium ions to capture the oxygen at the center, instead of the outside of the crystal. However, this apparatus requires new lasers which may result in a loss of image resolution, making it harder to identify our trapped oxygen.

Methods. We positioned a 1951 USAF resolution test target beneath the aperture of the Andor iXon 885 Imager and reflected light from various LED and laser sources beneath the target. We were mainly concerned with comparing the 313 nm laser for exciting Beryllium ($S_{1/2} \rightarrow P_{1/2}$) to the 397 nm laser for exciting Calcium. We used a 395 nm LED instead for ease of this test, however we intend to use a 397 nm laser in the actual trap.

Results. At both wavelengths, we are able to image stripes of width and distance apart $2.2\mu\text{m}$ from Group 7, Element 6 on the test target. Using a rectangular integrating brightness function across these stripes we are able to construct a brightness vs distance graph to identify the stripes by the peaks separated by troughs. While the range of gray value from trough to peak was around 3,000 to 12,000 on average for the 313 nm laser, the range for the 395 nm LED was 3,000 to 4,500 on average. Furthermore a qualitative inspection concluded that the test on the 395 nm LED appeared more indistinct, yet the lines were still distinguishable.

Conclusion. The lower range and lower average gray value for the 395 nm LED can likely be attributed to the less coherent light from the LED and a spherical aberration in the imaging system's lenses, which result in a slightly blurry image. While we observed these measurement differences for stripes a few micrometers apart, identifying ions spaced less than $10\mu\text{m}$ apart is still possible when using Calcium, with only a slight reduction in resolution. The difference is noticeable but will not significantly affect the functionality of our experiment.

Avoided-Crossing Coupling and Control of DPPH Spins On-Chip at Cryogenic Temperatures

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Purpose. Silicon spin quantum dots are a promising quantum computing paradigm due to their scalability and compatibility with widespread industrial fabrication techniques. The standard method of controlling these quantum dots is through pulsed microwave signals, but integrating these signals into the quantum dot environment poses significant challenges. Coplanar waveguides (CPWs) are a chip architecture that allows electrical components to be written onto the same chip as a silicon spin quantum dot, providing a path to interfacing a quantum dot with a larger apparatus. Additionally, prior research has shown the quantum effect of avoided crossing to be a viable means of coupling a spin system to a power feed line. We aim to create a CPW component with the potential to couple to silicon spin quantum dots via avoided crossing and integrate with other components in a quantum computational apparatus, such as a superconducting Mach-Zehnder amplitude modulator (AM).

Methods. We designed a component where an electrical feed line is capacitively coupled to hangar resonators of various sizes which connect to the CPW ground plane at one end and etched this design into Niobium deposited onto a Silicon wafer. We tested avoided crossing capabilities by applying DPPH, a standard material in electron spin experiments and therefore a good substitute for quantum dots. We applied DPPH to the resonators in two ways: suspended in isopropanol or in vacuum grease

Results. From measuring the frequency spectrum of the circuit as a function of magnetic field, we found that all four resonators which had vacuum grease applied to them experienced a noticeable effect when the cavity and DPPH spin systems were on-resonance, but only two of them displayed the expected mode-splitting behavior caused by avoided crossing. Further testing showed that one of the fully coupling cavities is within the 100 MHz tunability range of a mode of an AM connected in series.

Conclusion. . These results suggest the viability of CPW hangar resonators for quantum control applications and allow our group to proceed with testing the amplitude modulation component, which advances our long-term goal of producing low-dissipation quantum circuitry components.

Benchmarking the Coffea Framework at the Coffea-casa Analysis Facility

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Purpose. The volume of data generated by the High-Luminosity Large Hadron Collider (LHC) will be 100 times that of the current LHC, making dedicated computing facilities necessary for efficient data analysis. Coffea-casa is one such analysis facility, featuring an easy-to-use JupyterHub interface and access to scalable computing resources. Benchmarking the scalability of computing resources is essential to understand how to best utilize these resources for future physics analyses.

Methods. We run the full TopEFT analysis on 24 datasets, comprising over 80 million events, a total of 42 times. In each job, we primarily vary the number of workers to understand how this affects the analysis runtime. We also explore different data access protocols and versions of the coffea framework to understand their runtimes.

Results. We find no significant differences in performance when accessing data via XRootD with or without the XCache configuration, even though XCache is optimized for faster access. This is likely because the accessed data was stored at the University of Nebraska-Lincoln. We find that the analysis runs five times slower in the updated version of the coffea framework, as compared to its previous usable version.

Conclusion. –

Deriving Temperatures for Substellar Objects Within 20 Parsecs Using New Dynamical Masses

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Purpose. To deepen our understanding of the solar system and Earth's place in the galaxy, we must first obtain a sample of reliable spectra to investigate trends in the chemical compositions of substellar atmospheres. This can be achieved with direct imaging, which requires high-precision measurements of dynamical masses and orbital parameters. Using orvara, a Markov Chain Monte Carlo approach with state-of-the-art statistical techniques for efficient orbit solving, we derive, for the first time, complete system parameters for 29 objects in 25 systems: 21 stars, 2 brown dwarfs, and 6 planets.

Methods. By combining 5,000+ radial velocity measurements, 1,000+ relative astrometric observations from the Washington Double Star Catalog, and absolute astrometry from the Gaia DR3 version of the Hipparcos–Gaia Catalog of Accelerations, we calculate companion masses dynamically, improving upon previous model-dependent masses, as well as minimum masses that have inclination degeneracies. With these true masses and orbital parameters, notably semi-major axes, eccentricities, and inclinations, we constrain the three-dimensional architectures of stellar and substellar systems. Our new dynamical masses, paired with known host star ages, are used as input to substellar evolutionary models to derive the first-ever temperatures for five objects.

Results. Using substellar evolutionary models, we calculate new temperatures for 2 brown dwarfs: HIP 1292B (1330.3 K) and HIP 10812B (1119.4 K), as well as 3 exoplanets: HIP 1292Ab (381.8 K), HIP 71395b (249.6 K), and HIP 71395c (298.8 K). This establishes each as an exciting candidate for atmospheric characterization with JWST, which has a threshold of ~300 K for optimal direct imaging.

Conclusion. Our robust sample represents the newest and highest precision constraints on systems in the solar neighborhood and will set the stage for the upcoming Nancy Grace Roman Space Telescope and Habitable Worlds Observatory, which will be capable of directly imaging Earth-like exoplanets.

Galaxy Clusters as a Bridge between Theory and Observation

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Purpose. In current cosmology, CDM is considered as a standard model of cosmology that best explains the evolution of the universe with only six parameters. Dark energy (Λ) and cold dark matter (CDM) lie at the core of CDM, yet their identities still remain as a question. Constraining the cosmological parameters is crucial to understand the nature of dark energy, and galaxy clusters, which are formed from initial density fluctuations in the early universe, are considered as one of the most powerful cosmological probes. However, using clusters as a cosmological probe requires mass-richness relation ($P(M)$), since the cluster mass is not a direct observable. In this research, we build a pipeline that constrain parameters of $P(M)$ and seek ways to improve in a framework called full forward modeling.

Methods. In order to build a pipeline, we used a mock catalog of clusters from Sunayama et al. 2020. We used three observables to constrain the parameters of $P(M)$: abundance, stacked lensing signal, and clustering. On the other hand, we computed the observables from cosmological parameters and $P(M)$ using `dark_emulator`, a cosmological emulator developed in Nishimachi et al. 2019. To evaluate the parameters, we apply Bayesian analysis through the Markov chain Monte Carlo (MCMC), which collects “likely” parameter values by random-walking and returns a probability distribution of parameter values called posterior.

Results. From our best run of MCMC, most parameters are normally distributed and constrained well with small 1, yet one parameter shows two peaks in its probability distribution. When comparing $P(M)$, the predicted $P(M)$ is narrower in every bin. Potential causes include incorrectly assuming the same relationship across all z and mass bias from weak lensing. As future steps, we will seek ways to eliminate the slight disagreements with the sample and would like to add cosmological parameters as free parameters. We will then run the pipeline with the CAMIRA cluster catalog, possibly with multiple redshift values.

Conclusion. In this research, we build a pipeline that constrain parameters of $P(M)$ for future analysis with real cluster catalog and cosmological parameters. The results of MCMC mostly match the sample with a slight disagreement in width of posterior distribution. The parameter values also show slight disagreement from previous studies, yet the uncertainties became smaller with the presence of clustering in addition to abundance and lensing signal. In the future, we will run the pipeline with real data and incorporate cosmological parameters as free parameters.

Identifying Multiple-star Systems in the Ultracool Dwarf Sample

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Purpose. The lowest mass stars and brown dwarfs, also known as ultracool dwarfs (UCDs; objects less than ~15% the mass of the Sun), are a crucial population in the search for habitable worlds. However, exoplanet population statistics have yet to be well studied among UCDs because of inefficient exoplanet detection methods around these faint objects, resulting in the need for target prioritization for the most probable exoplanet hosts. A potential indicator of exoplanet host status is a spectroscopic peculiarity shared by TRAPPIST-1 and Teegarden's star, both of which are exoplanet hosts with at least two planets in their habitable zones.

Methods. A sample of over 800 UCDs has been constructed, and their low-resolution, near-infrared spectra have been collected over the past 5 years. This sample is key to testing the potential correlation between the spectral peculiarity in the UCDs and the existence of planetary systems. However, first we must separate the multi-star systems from the single-star systems to maintain statistical robustness. We use the Gaia Archive and Washington Double Star Catalogue to identify UCDs in our sample that are part of multiple-star systems.

Results. We found that 93 of our target UCDs are a part of multiple-star systems, 17 being triple or higher degree systems, and 76 being binary systems.

Conclusion. These multi-star systems are interesting targets to follow up with and test if the stars in the system also show this spectral peculiarity, therefore potentially being multi-star systems hosting exoplanets. Additionally, since ~50% of all Sun-like stars are in binary systems (a multi-star system consisting of two stars), and planets have been discovered in binary systems (circumbinary planets), these systems are essential to understand, especially for upcoming missions like the Habitable Worlds Observatory, which will look at planets in binary systems.

Let's Get Metal: An Analysis of Gas-Phase Metallicity for Galaxies within the DIVE Survey

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Purpose. This analysis looks at the metal abundances of dwarf galaxies living in void environments. Using Integral Field Unit (IFU) data, this analysis aims to develop a new method of measuring metal in a galaxies' gas and utilize different calibration methods in order to discover possible trends in a Gas-Phase Metallicity (GPM) plot.

Methods. This analysis used the data obtained from de los Reyes et al. (2023) through the KCWI spectrograph. Spectral data were gathered through a method of weighted integration which allows the entire galaxy to be included. Using the documented and appropriate GPM calibration methods outlined in Kewley & Ellison (2008), the appropriated spectral lines indicating certain elements within the gas were able to be isolated. The lines were then fitted with a gaussian distribution and integrated in order to measure the total flux. With this flux, their GPM's were calculated and plotted. Due to a lack of data (5 control galaxies, 16 void galaxies), no statistical analysis was performed.

Results. The results of this analysis show that IFU data have promising uses on the further study of intergalactic objects, especially galaxies. This is one of the first times full-body representation has been considered in looking at galaxy spectra and can provide more accurate analysis. As can be seen in the final GPM plots, across all calibrations, none of the DIVE galaxies seem to indicate a trend. While this does not rule that a trend won't emerge upon the collection of more data, it does validate the accuracy of spectral integration. Further research would include the analysis of dwarf galaxies to find statistical significance of any trends, and the analysis of large galaxies to confirm if spectral integration in IFU data matches known GPM trends.

Conclusion. In conclusion, this analysis provides strong evidence for the accuracy and use of IFU data in spectral analysis of objects. The similarities between the GPM plots suggests a consistent measurement of metals with galaxies. This validates the use and need for more images under the IFU format.

Monte Carlo Simulations & Phase Transitions

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Purpose. In this study we investigate Monte Carlo methods and phase transitions through random walks and lattice percolation, and how to utilize the Metropolis and Worm algorithms to calculate observables for the 2D Ising model. The random walks and percolation are primarily done to learn about Monte Carlo methods and phase transitions, focusing on observable such as mean squared displacement as a function of step count, and various observables for lattice percolation and clusters.

Methods. When looking at the 2D Ising model we measure various observables with both Metropolis and Worm algorithms. We then compare their values to verify the functionality of our Worm algorithm.

Results. The dynamical critical exponent is calculated for both algorithms: Metropolis: $z_{\text{metro}} = 1.856 \pm 0.063$, Worm: $z_{\text{worm}} = 0.833 \pm 0.030$.

Conclusion. We find the worm algorithms functions more efficiently and avoids critical slowing down unlike the metropolis algorithm.

Noise: Mitigation of Uncertainty in the Precision Measurement of Long-Range Spin-Dependent Interactions

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Purpose. Long-range spin-dependent interactions (LRSDIs) are forces predicted by some proposed extensions to the Standard Model. We investigate the theorized effect by measuring the Larmor precession frequency of spin-polarized cesium-133 (^{133}Cs) and mercury-199 (^{199}Hg), composing our signal. A signal is the raw electronic data generated by optical detectors receiving electromagnetic input. Noise, in contrast, is anything that obscures the signal. While random error, uncertainty introduced by unpredictable factors, has largely been minimized, systematic error introduced by recurring discrepancies shifting the data in a consistent data is more concerning. We correct for known systematic deviations in our data and assess the remaining uncertainties associated with our correction. This summer's work investigated various factors to further identify and minimize sources of error.

Methods. Taking data in multiple “configurations,” where parameters are altered, alerts us to systematic effects. We degauss between these different configurations to get rid of potential lingering systematic effects associated with the magnetic shields. To assess the impact of parameter variation on the precession frequencies, we measure the Hg-Cs difference field versus the slope of potential variables. We can measure the deviation of new data from historical saga data using two-sigma uncertainty analysis to identify outliers. We subtract systematic effects that we cannot physically eliminate from our final calculation of the Hg-Cs difference field.

Results. We made several changes decreasing experimental noise: offsetting the Hg probe laser trigger to align the phase with the trough of Hg frequency best-fit error, cutting the first four measured points of observed Cs frequency from every cycle, and limiting run times to periods without high seismic noise. We shifted the phase of the electronic feedback within the 2nd harmonic generator, which improved our laser stability and potentially increased the tolerance of Hg error against construction-related noise. Additionally, we isolated the gradient coil current supply from our ground, eliminating the possibility of a ground loop.

Conclusion. This summer, with our primary goals being data collection and error reduction, we made significant improvements to our uncertainty and error metrics. There remain several issues, such as humidity fluctuations that correlate with the Cs field gradient, the inability to suppress third-order or higher effects in time, and lifetime decay of the cells. We plan to take data at understudied configurations to identify potential systematic errors, continue to take additional data to further improve statistical error, and run diagnostic checks on existing data.

Signals: Searching for Long-Range Spin-Dependent Interactions

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Purpose. Theorized extensions of the Standard Model predict an additional, spin-dependent fifth fundamental force. Our experiment uses the Earth as a source of spin-polarized electrons, and an ^{199}Hg - ^{133}Cs co-magnetometer as a detector to look for long-range candidate particles for that force. The electron spin in Cs has been shown to not exhibit any exotic behavior, so we use the Cs signal as a baseline for the local magnetic fields and their associated effects. This leaves the nuclear spin of ^{199}Hg and its associated valence neutron sensitive to exotic spin-dependent forces. The units of our measurement are Hz of Hg precession frequency (Hg Hz).

Methods. Our set-up uses five magnetic field coils to create a homogeneous magnetic field within the apparatus, and an additional, sixth coil to cancel out Earth's field. The Hg and Cs atoms are first pumped up with circularly polarized light tuned to their absorption frequency and intensity modulated at their Larmor precession frequency. They are then probed with linearly polarized light tuned away from their absorption frequency. A rotating table in our apparatus aligns the precession field of the atoms with north, south, west and east to probe the different alignments to dot or cross product-dependent candidate interactions. The North-South and East-West signals are then subtracted to get our final result. Our data is collected into octets, EW or NS, consisting of 8 distinct table positions ordered as ABBABAAB to allow us to subtract off first and second order drifts. The octets are then divided into sagas based on their time sequence and magnetic field orientation. The direction of the precession fields is flipped for half of the data to measure and cancel out systematic effects related to table position.

Results. A null result in NS is 17.112 Hg Hz and 0.000 Hg Hz in EW. We found results of $-0.53 \pm 0.056 \text{ stat} \pm 0.04 \text{ syst Hz}$ in EW and $17.72 \pm 0.055 \text{ stat} \pm 0.04 \text{ syst Hz}$ in NS.

Conclusion. We achieved our aim of 0.07 Hg Hz of systematic and statistical uncertainty. This experiment will place bounds on these candidate interactions one order of magnitude better than our laboratory's previous experiment.

Psychology

Making Intervention Do-ABLE: Analysis of Police Officers' Perceptions of Active Bystandership Training

Jiaqi Huang (1), Viktoria E. Leopold (1), Cassandra Ramdath (2), Catherine A. Sanderson (1,2)

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Purpose. Purpose: This study investigates large-scale officer feedback on what agencies, direct supervisors, and each individual officer should do to improve active bystandership. This will serve to fix the research gap on how officers themselves perceive the effectiveness of ABLE training, and what they believe can strengthen its impact.

Methods. Methods: This qualitative study analyzed responses from 48,077 officers across over 400 police departments in the USA and Canada to an open-ended national survey that explored perceptions, attitudes, and beliefs about active bystandership and peer intervention. The survey was administered following an 8-hour virtual officer training program. Researchers analyzed each question thematically based on an established codebook, tested for interrater reliability, and used it to identify patterns and themes related to how individual officers, supervisors, and agencies can support active bystandership.

Results. Results: To promote active bystandership, officers stressed ongoing training along with real-world application, and though most found ABLE effective or already in use, some suggested shorter, in-person training. They emphasized open discussion across all ranks, reflections after incidents, regular reminders to reinforce intervention, and outreach via promotions to share ABLE with other communities. Officers urged leadership to verbally support active bystandership through exemplary behavior, direct field involvement, and called for accountability. They also asked for recognition of successful interventions through formal or verbal awards. Officers called for mandatory ABLE training and policies with firm anti-retaliation protections, expressing that confidence in intervention increases when officers are shielded from backlash and negative consequences. They stressed the importance of individual action through speaking up and intervening, openness to feedback across all ranks, and continuous personal growth to build confidence and intervention skills. Officers want a culture of awareness and supportive relationships that promote intervention while addressing mental and physical well-being challenges created by toxic leadership, staffing shortages, and negative work environments.

Conclusion. Conclusion: Departments can use this information to refine ABLE's format, while also implementing the policies that support intervention, reducing police violence and misconduct. Promoting ABLE and sharing successful stories also fosters accountability to rebuild public trust in law enforcement. Future research could examine connections between demographic data and officers' responses and recommendations for encouraging ABLE.

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Understanding Mania: The Role of Emotion Beliefs and Emotion Regulation in Mania Symptoms

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Purpose. This study explores the influence of emotion beliefs and emotion regulation strategies on mania symptoms. Previous research indicates that viewing emotions as fixed, unique, or long-lasting is linked to greater psychopathology, and maladaptive cognitive strategies specifically increase risk for mania. By exploring the role these factors have in mania symptoms, the study aims to improve our understanding of mania and offer insights that could inform effective approaches to managing mania symptoms.

Methods. In this study, 186 participants (66.1% women; mean age = 28; 61.9% White/Caucasian) completed several questionnaires through Qualtrics that assessed their baseline mania symptoms, general emotion beliefs, and emotion regulation strategies. The latter two were then reassessed through one-month follow-up surveys. Since the study examines mania specifically, only data from 41 participants who reported elevated mania symptoms indicative of a highly probable manic or hypomanic condition were included in the analysis.

Results. Baseline correlation analysis revealed that mania symptom severity is positively associated with maladaptive regulation strategies such as brooding rumination, catastrophizing, and beliefs that emotions are unique and long-lasting. Interestingly, mania symptom severity is not significantly associated with emotion malleability beliefs. Further analysis into brooding rumination revealed that brooding thoughts of feeling wronged, self-criticism, or perceiving one's experiences as unique are linked to greater mania severity. At the one-month follow-up, longitudinal analysis indicated that baseline mania severity was positively correlated with stronger brooding thoughts of self-blame and perceiving experiences as unique.

Conclusion. This study concludes that certain negative emotion beliefs and maladaptive regulation strategies are linked to greater mania symptom severity. Contrary to expectations, emotion malleability beliefs are not significantly associated with mania symptoms. These findings improve our understanding of mania and offer insights toward developing more effective strategies in managing mania symptoms. Future research could expand these findings with a larger, clinical sample and a longitudinal design that measures mania severity at multiple time points.

STEM Education

Impact of Structured Mentoring Practices on Undergraduate Summer Research Experience

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Purpose. Effective mentorship is essential for student success in undergraduate research experiences. However, specific mentoring practices that have been shown to be effective are currently under-researched, particularly in STEM programs. The Facilitating Undergraduate Research Through Holistic and Extensive Skills Training (FURTHEST) project seeks to explore how undergraduate research experiences in STEM are shaped by structured mentoring practices.

Methods. We employed a mixed methods approach, including student interviews and surveys, to examine the efficacy of two mentor practices in particular, daily logs and an intensive "bootcamp" week.

Results. After reading through students' interview transcripts and daily logs, bucket codes were developed for Bootcamp and Daily Log mentions and were then consolidated into emerging themes, which were used throughout the qualitative coding process to gather further insights about student experience. Next steps will include analyzing the coded data, exploring the impact of other structured mentoring practices, and developing conclusions about effective mentoring practices.

Conclusion. This research is essential for establishing effective mentoring practices that can improve student growth in undergraduate research programs.